

1. MoistureMapper: An Autonomous Mobile Robot for High-Resolution Soil Moisture Mapping at Scale (2025)

Year: 2025

Publisher: IEEE CASE 2025

Research Area: AI-Based Smart Agriculture and Autonomous Farming System

Why it is Relevant:

Connects AI-based decision-making with smart agriculture and autonomous farming applications. The system uses autonomous robotics and intelligent adaptive sampling to collect soil moisture data for precision irrigation and farm management.

Summary:

The paper presents MoistureMapper, an autonomous ground robot equipped with TDR sensors for high-resolution soil moisture mapping. The robot autonomously navigates agricultural fields, collects soil moisture measurements, and generates moisture distribution maps using Gaussian Process-based adaptive sampling algorithms. Experimental results show improved mapping performance and reduced travel distance compared to benchmark approaches. The system supports precision agriculture through efficient irrigation planning and water management.

Relation to Your Project:

- Autonomous farming robot ✓
- Soil moisture monitoring ✓
- Precision irrigation ✓
- AI-based adaptive decision making ✓
- Real-time agricultural data collection ✓
- Smart agriculture automation ✓